Assignment 1

1) Program to print the Even Number till 50 (with Tracing of Loop)

→ print("Even numbers from 1 to 50 :")

```
for i in range(1, 51):
    if i % 2 == 0:
        print(i)
```

Tracing of Loop

Iteration	i	i % 2	condition i % 2 == 0	output
1	1	1	False	-
2	2	0	True	2
3	3	1	False	-
4	4	0	True	4
5	5	1	False	-
6	6	0	True	6
...	...	...	...	...
49	49	1	False	-
50	50	0	True	50

Logic :- A number is even when its remainder after division by 2 is 0.

2) Program to print the Prime Numbers till 100 (with Tracing of Loop)

→
```
print("Prime numbers from 1 to 100:")

for num in range(2, 101):
    count = 0

    for i in range(1, num + 1):
        if num % i == 0:
            count = count + 1
    if count == 2:
        print(num)
```

Tracing of Loop

num	Number of Factors	Prime ?
2	2	Yes
3	2	Yes
4	3	No
5	2	Yes
6	4	No
7	2	Yes
8	4	No
9	3	No
10	4	No
11	2	Yes

The same process continues until num = 100.